9.7 Power Series

From infinite series of numbers, we can study "infinite polynomials" - **power series**.

Definition: A **power series about** $x = 0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \ldots + c_n x^n + \ldots.$$

A **power series about** $x = a$ is a series of the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1 (x - a) + c_2 (x - a)^2 + \ldots + c_n (x - a)^n + \ldots$$

in which the **center** a and the coefficients c_o, c_1, c_2, ..., c_n, ... are constants.

Example: The geometric power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots + x^n + \ldots$$

converges to $1/(1 - x)$ for $|x| < 1$. This is usually express by writing

$$\frac{1}{1-x} = 1 + x + x^2 + \ldots + x^n + \ldots, \quad -1 < x < 1.$$

Note: The partial sums of the series are the polynomials $P_n(x)$ that approximate the function on the left. The approximations do not apply when $x \geq 1$.

Example: Consider the power series

$$1 - \frac{1}{2}(x - 2) + \frac{1}{4}(x - 2)^2 + \ldots + (-\frac{1}{2})^n(x - 2)^n + \ldots$$

We can test the convergence of a power series using the Ratio Test.

Example: For what value of x do the following power series converge?

(a). $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

(b). $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$

(c). $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$

(d). $\sum_{n=1}^{\infty} n!\, x^n = 1 + x + 2!\, x^2 + 3!\, x^3 + \ldots$

Theorem (The Convergence Theorem for Power Series):

If the power series $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \ldots$ converges at $x = c \neq 0$,then it converges absolutely for all x with $|x| < |c|$. If the series diverges at $x = d$, then it diverges for all x with $|x| > |d|$.

Proof:

The Radius of Convergence of a Power Series

Corollary: The convergence of the series $\sum c_n(x - a)^n$ is described by one of the following three cases:

1. There is a positive number R such that the series diverges for x with $|x - a| > R$ but converges absolutely for x with $|x - a| < R$. The series may or may not converge at either of the endpoints $x = a - R$ and $x = a + R$.

2. The series converges absolutely for every x ($R = \infty$).

3. The series converges at $x = a$ and diverges elsewhere ($R = 0$).

R is called the **radius of convergence** of the power series, and the interval of radius centered at $x = a$ is called the **interval of convergence**.

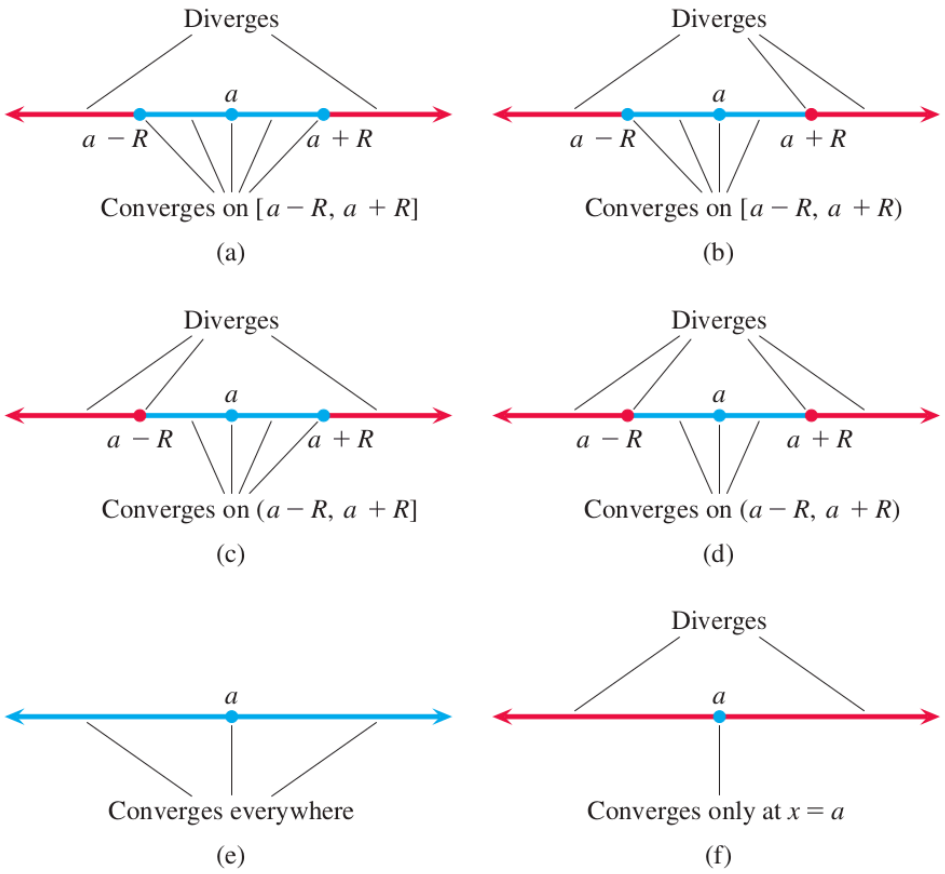


FIGURE 9.21 The six possibilities for an interval of convergence.

How to Test a Power Series for Convergence

1. *Use the Ratio Test (or Root Test) to find the interval where the series converges absolutely.* Usually, this is an open interval

$$|x - a| < R \qquad \text{or} \qquad a - R < x < a + R.$$

2. *If the interval of absolute convergence is finite, test for convergence or divergence at each endpoint.* Use a Comparison Test, the Integral Test, or the Alternating Test.

3. *If the interval of absolute convergence is* $a - R < x < a + R$*, the series diverges for* $|x - a| > R$ (it does not even converge conditionally) because the nth term does not approach zero for those values of x.

Examples:

1. $\sum_{n=0}^{\infty} \frac{(x-2)^n}{10^n}$

2. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x+2)^n}{n2^n}$

Operations on Power Series

Theorem (The Series Multiplication Theorem for Power Series): If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$ converge absolutely for $|x| < R$, and

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \ldots + a_{n-1} b_1 + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k}$$

then $\sum_{n=0}^{\infty} c_n x^n$ converges absolutely to $A(x)B(x)$ for $|x| < R$:

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n.$$

We can also substitute a function $f(x)$ for x in a convergent power series.

Theorem: If $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for $|x| < R$, then $\sum_{n=0}^{\infty} a_n (f(x))^n$ converges absolutely for any continuous function f on $|f(x)| < R$.

Example: $1/(1 - 4x^2) = \sum_{n=0}^{\infty} (4x^2)^n$

Theorem (The Term-by-Term Differentiation Theorem): If $\sum c_n(x-a)^n$ has radius of convergence $R > 0$, it defines a function

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n \text{ on the interval } a - R < x < a + R.$$

The function f has the derivatives of all orders inside the interval, and we obtain the derivatives by differentiating the original series term by term:

$$f'(x) = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1},$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1)c_n(x-a)^{n-2},$$

and so on, Each of these derived series converges at every point of the interval $a - R < x < a + R$.

Example: Find series for $f'(x)$ and $f''(x)$ if

$f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots, \quad -1 < x < 1.$

A power series can also be integrated term by term throughout its interval of convergence.

Theorem (The Term-by-Term Integration Theorem): Suppose that $f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n$ converges for $a - R < x < a + R \quad (R > 0)$. Then

$$\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

converges for $a - R < x < a + R$ and

$$\int f(x)\, dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C$$

for $a - R < x < a + R$.

Example: Identify the function

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots, \quad -1 \le x \le 1.$$

Example: Consider the series

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + \ldots$$